Mega XL — custom upgrades & build notes

Upgrades to a Kickstarter v1.0 Mega XL CNC router.

This documents the changes made to a first-generation (Kickstarter v1.0) Mega XL: a taller, stiffer Z, closed-loop motors, all-proximity homing, and a complete electronics rebuild around a grblHAL controller in a custom enclosure with a proper 48 V power system, air assist, and probing.

At a glance

Subsystem	Stock (v1.0)	Upgraded
Vertical clearance	stock	+3″ via Mark Smith riser kit
Z assembly	stock Z	Mark Smith Z gantry (only V-wheels + X motor mount reused)
Spindle	DeWalt DWP611 (1¼ HP router)	2.2 kW (3 HP) Φ80 mm water-cooled, ER20, 24k RPM, VFD-driven
Stepper motors	generic open-loop steppers	all-in-one closed-loop steppers (driver integrated in motor)
Limit / homing	mechanical limit switches	proximity sensors on every axis
Probing	—	touch plate + 3D probe in parallel; grounded spindle (no tool clip)
Controller	8-bit Arduino board	grblHAL Teensy 4.x (USB + Ethernet)
Electronics enclosure	cramped stock steel box	custom 12×16×6″ aluminum / plexiglass
Power	stock	48 V Meanwell system, E-stop, buck-derived 12 V / 5 V rails
Air assist	none	Mist-relay 12 V solenoid → 30 PSI nozzle at the cut
Drag chains	stock 1.5	Y axis upsized to 2″ to fit the full cable bundle
Dust collection	—	4″ swivel + 5′ flex to a spindle dust shoe, on an overhead reel

Frame & mechanical

- **Mark Smith riser kit** — adds **3" of additional vertical (Z) clearance**, allowing taller stock and longer tooling.
- **Mark Smith Z gantry** — a **complete replacement of the Z assembly**. The only stock parts carried over are the **four V-wheels** and the **X stepper motor mount**; everything else is new.

Spindle & cooling

The stock **DeWalt DWP611** (1¼ HP trim router) was replaced with a **2.2 kW (3 HP) water-cooled spindle** kit:

- **Spindle** — Φ80 mm body, 220 V, water-cooled, **ER20 collet**, up to **24,000 RPM at 400 Hz**.
- **VFD** — 2.2 kW (3 HP) inverter, fed from the **single-phase 220 V** circuit through the E-stop; it generates the **3-phase output to the spindle** and is commanded by the breakout board's **0–10 V PWM** speed signal (see Power distribution).
- **Mount** — 80 mm spindle clamp on the new Z gantry.
- **Braking resistor** — a 150 W resistor on the VFD dissipates regen energy for a fast stop (spindle down in **under 6 s**).

Coolant loop. A pump **submerged in a 1-gallon reservoir** circulates coolant through the spindle via **two 8 mm tubes** (supply and return): *reservoir → pump → spindle → radiator → reservoir*. The radiator is cooled by the 12 V fan (see the power tree). The pump is **interlocked to the spindle** — a **VFD digital output** drives the **NO contact of a 110 V relay** the pump plugs into — so it runs whenever the spindle does and the spindle can't be run dry. (Separate from the breakout board's Mist relay, which handles air assist.) The coolant tubes, the air-assist line, and dust extraction all reach the spindle through the overhead umbilical (below).

Motors

The generic steppers that shipped with the Mega XL were replaced with **all-in-one closed-loop stepper motors** — each motor has its stepper driver built into the motor body. This removes the separate driver boards, adds closed-loop position feedback (lost-step detection / alarm), and simplifies wiring to a single cable per motor.

- **Individual enable** per motor — each stepper's `EN` line is driven separately.
- The motors' `AL` **(alarm) outputs are wire-OR'd together** and fed to the controller's `ST0` **input**, so a stall / lost-step fault on any motor signals the controller.

All four axes are closed-loop. The machine is a **dual-Y gantry** — the two Y gantries are driven by ganged motors (**Y + A**), one per side.

Axis	Holding torque	Drives
Y + A	3.2 N·m each	the two Y gantries (ganged)
X	2.0 N·m	spindle carriage along the cross-beam (rack & pinion)
Z	1.2 N·m	Z

The breakout board also provides a **5th axis (B)**. No rotary is fitted yet, but its **step/dir/enable is wired through to the extension panel** so one can be added later by plugging it in — see [Back-of-enclosure harness & connectors].

All motors have an **8 mm output shaft with a flat**, with pinions/couplings retained by a grub screw on the flat. *Build note:* the stock Mega XL pinions were 6 mm-bore and were **reamed to 8 mm** at a machine shop — this thinned the set-screw boss and reduced grub-screw thread engagement. See the [rack-and-pinion maintenance] appendix.

Homing & limit sensing

The Mark Smith Z gantry uses a **proximity sensor** as its limit switch. To keep one sensing scheme across the machine, **all remaining stock mechanical limit switches were replaced with the same style of proximity sensor**, so every axis homes on identical, non-contact hardware.

Probing

- The **touch-plate input is wired in parallel with the primary probe input**, alongside the **3D mechanical probe** — either device can trigger the same probe input.
- The **spindle mount is grounded**, so the tool itself is at ground through the spindle. That completes the touch-plate circuit on contact — **no alligator clip on the tool** is required.

Controller & electronics enclosure

- Discarded the **cramped stock steel electronics enclosure** and the **stock 8-bit Arduino controller board**.
- Control is now a **grblHAL Teensy 4.x** controller on a breakout board (32-bit).
- Built a **custom 12″ × 16″ × 6″ enclosure**:
  - Frame: **2020 aluminum extrusion**.
  - Five sides: **0.0625″ (1/16″) aluminum sheet**.
  - Top: **1/8″ clear plexiglass** (visibility into the electronics).

Connectivity

The Teensy 4.x supports Ethernet, and the breakout board has the **Ethernet option installed**. Both links to the Teensy are brought out as **panel patch-through connectors on the side of the enclosure**: a **female USB-A** jack and a **female Ethernet (RJ45)** jack.

Power distribution

A single **220 V, 20 A branch circuit** feeds the machine through a **front-panel emergency-stop button** (the E-stop breaks the 220 V input, so it drops both the PSU and the spindle drive). Building power arrives as **12/3 from the breaker panel to a junction box**, then via an **extension-cord pigtail** to the panel C20 and the E-stop input; a second pigtail carries the E-stop output to the VFD. This is **single-phase 220 V** (residential); the VFD generates the 3-phase output to the spindle. The 220 V powers an **800 W 48 V Meanwell power supply** and the **VFD**. Everything else is derived from the 48 V rail through two buck converters.

220 V, 20 A → front-panel E-stop →

└─ 48 V Meanwell power supply

└─ VFD (spindle drive)

48 V (PSU output) →

└─ 4 × stepper motors

└─ 12 V / 3 A buck converter

└─ 5 V / 3 A buck converter

12 V (buck) →

└─ 4 × proximity limit sensors

└─ 12 V fan cooling the spindle coolant loop (runs while the spindle runs)

└─ grblHAL Teensy 4.x breakout board, 12 V input — the board uses 12 V to generate the **0–10 V PWM spindle-speed signal** to the VFD

5 V (buck) →

└─ breakout board 5 V input — board logic power, **separate from the Teensy** (the Teensy is powered over USB)

└─ 3D probe

Relay outputs & air assist

The breakout board provides **5 relay outputs**, each **jumper-selectable for 5 V or 12 V**. Only the **Mist** output is in use, driving the air-assist nozzle:

Mist relay → 12 V **NC** solenoid valve → regulator @ 30 PSI → 1/4″ tube (up the umbilical) → needle valve at top of Z gantry → 1/8″ copper airline → swivel nozzle aimed where the tool meets the stock

Connectors & terminations

Ferrules are crimped on all terminal-block connections — every stranded conductor landing on a terminal block is ferruled for a clean, captured, reliable termination.

Enclosure pass-throughs use **Molex Mini-Fit Jr panel-mount connectors**. The choice was deliberate: with roughly 60 terminations to make, the builder had far more confidence producing **~60 quality crimps** than ~60 quality solder joints on aircraft-style connectors.

Cable & wiring schedule

All runs use shielded cable.

Cable	Run	Conductors
18/6 shielded	enclosure → each stepper motor	6 (assignment depends on motor type — see below)
22/3 shielded	enclosure → Y-axis proximity switches	`12V`, `Sig`, `GND`
22/6 shielded	enclosure → X & Z proximity sensors + 3D probe (shared)	`12V`, `GND`, `XLim`, `ZLim`, `P`, `5V`
Spindle lead	VFD → spindle (3-phase, with the spindle kit)	`U`, `V`, `W`
22/4 shielded	breakout → VFD (control)	`0–10V`, run, GND, spare

18/6 stepper cable — conductor assignment

Conductor	Closed-loop stepper	Discrete (open-loop) stepper
1	`48V`	`A+`
2	`GND`	`A-`
3	`DIR`	`B+`
4	`PUL`	`B-`
5	`EN`	`EN`
6	`AL`	`AL`

All four axes on this build are closed-loop, so they all use the left-column assignment (`48V` / `GND` / `DIR` / `PUL` / `EN` / `AL`). `DIR` / `PUL` are direction/step to the integrated drives; `EN` = enable, `AL` = alarm. The discrete `A±` / `B±` column is kept for reference only — the same 18/6 cable would serve an open-loop motor if one were ever fitted. At the enclosure these device cables collapse onto a small set of keyed panel connectors — see Back-of-enclosure harness & connectors.

Drag chains (cable carriers)

Each axis's drag chain carries every cable that has to continue past it, so loading is cumulative toward the base of the motion chain. **Y is the fullest** — everything routes through the Y carrier first to reach the gantry, then the X carrier feeds the spindle carriage, while the opposite gantry rail (**A**) carries only its own stepper. The two limit-sensor drops that don't need to flex (A and Y) **begin after** their drag chains rather than riding through them.

Drag chain	Cables carried
A axis (opposite gantry rail)	1 × 18/6 (A stepper) *A limit (22/3) picks up after the chain*
X axis (spindle carriage)	1 × spindle lead (3-phase) 2 × 18/6 (X & Z steppers) 1 × 22/6 (X & Z limits + 3D probe)
Y axis (gantry — carries everything)	1 × spindle lead (3-phase) 3 × 18/6 (X, Z & Y steppers) 1 × 22/6 (X & Z limits + 3D probe) *Y limit (22/3) picks up after the chain*

That Y bundle — the spindle lead, three 18/6, and one 22/6 — would not fit the stock **1.5** drag chain, so the Y axis was moved up to the **next size larger (2")** carrier.

Back-of-enclosure harness & connectors

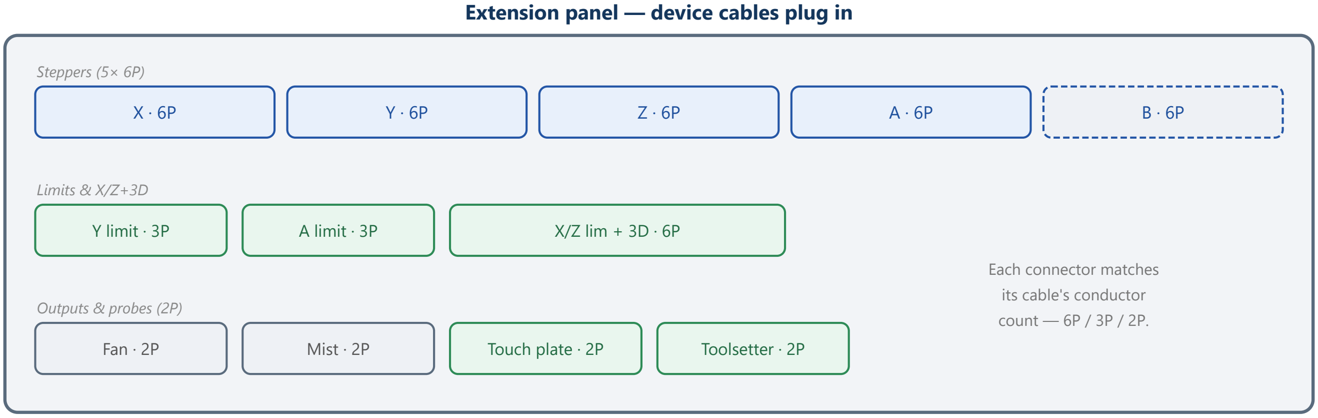
The wiring uses **two panels**. The **extension panel** is where the machine's device cables plug in — one socket per cable, sized to its conductor count. The **back panel** on the rear of the enclosure carries the collapsed result to the controller. Because the enclosure lives in a **cabinet drawer** and the back panel is connected **by feel** (drawer pulled forward, hand reaching in), the harness is squeezed down to just **five single-row 1×6 Mini-Fit Jr sockets** there — few connectors, and a linear pin row is easy to count and seat blind.

Extension panel — device cables plug in

Twelve sockets, one per device cable (50 conductors in):

Socket	Size	Cable	Conductors
Stepper X / Y / Z / A / B	5× 6P	18/6	`48V · GND · DIR · PUL · EN · AL`
Y limit	3P	22/3	`12V · Sig · GND`
A limit	3P	22/3	`12V · Sig · GND`
X/Z limits + 3D probe	6P	22/6	`12V · GND · XL · ZL · 3D · 5V`
Fan	2P	—	`12V · GND`
Mist	2P	—	`Mist · GND`
Touch plate	2P	—	`TP · GND`
Toolsetter	2P	—	`TS · GND`

Extension panel — device cables plug in



Behind the panels — the remap

Between the two panels, each incoming conductor is routed to its designated back-panel socket and pin. The signals that tie together — the **common ground**, the five `AL` **alarms wire-OR'd to one**

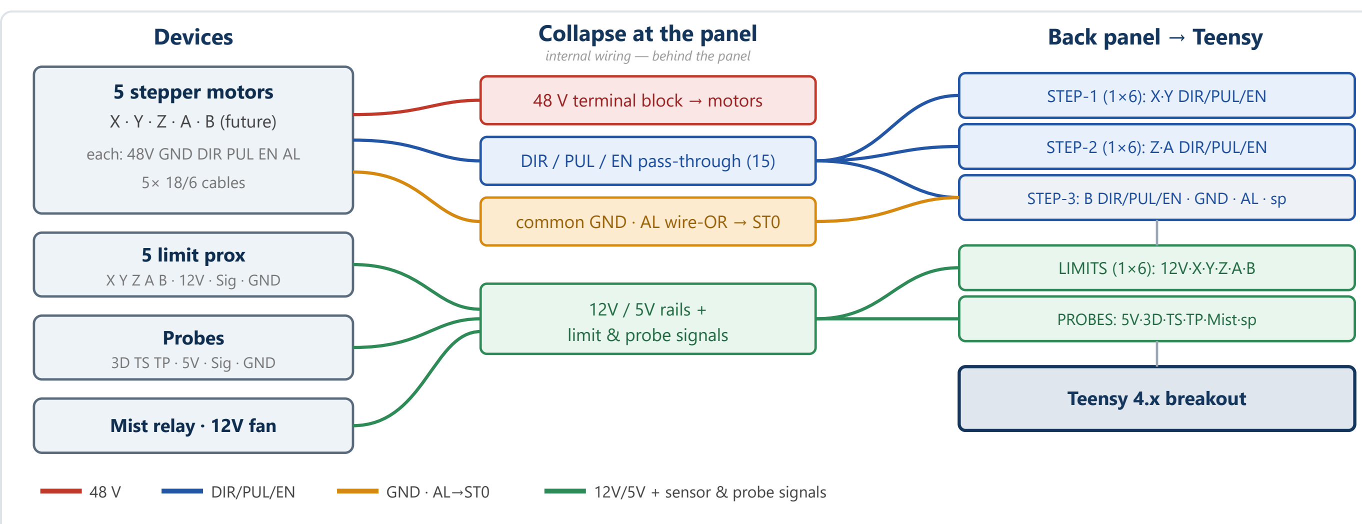
`ST0`, and the shared **12 V / 5 V** rails — are joined by **daisy-chaining from the incoming crimp terminals** (a jumper from one crimp to the next, not a bus bar). Each motor's **48 V splits off to a distribution terminal block**; the spindle **VFD has its own C20 220 V inlet** and a **22/4 control cable** from the breakout (0–10 V speed + run).

Back panel — to the controller

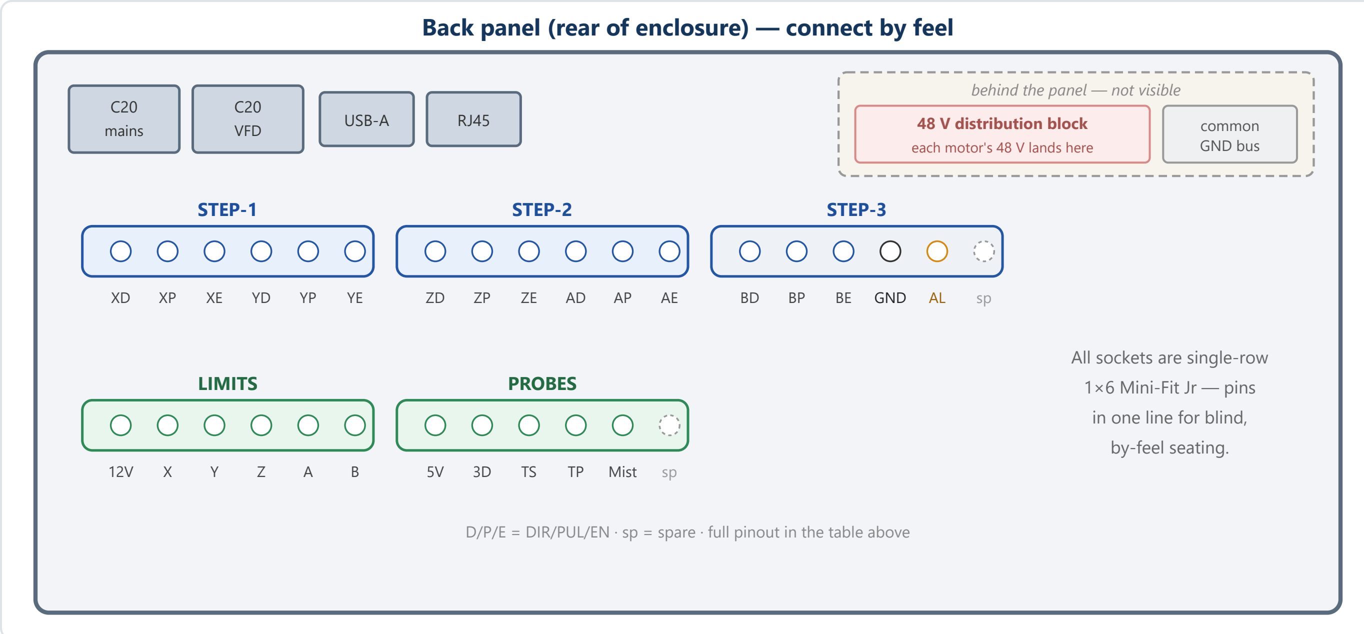
The collapse: a closed-loop motor only needs **DIR, PUL, EN** unique (48 V is its own rail, GND is shared, AL is wire-OR'd), so the 5 stepper channels go from 30 conductors to **5×3 = 15 + GND + AL = 17 → three 1×6 sockets** (one spare pin). The sensors and probes fill **two more 1×6 sockets**. Net: **five sockets / five cables** to the controller (three 18/6, two 22/6); the **48 V terminal block and common-ground bus are behind the panel**, not on the connect-by-feel face.

Socket (1×6)	Cable	1	2	3	4	5	6
Stepper 1	18/6	X DIR	X PUL	X EN	Y DIR	Y PUL	Y EN
Stepper 2	18/6	Z DIR	Z PUL	Z EN	A DIR	A PUL	A EN
Stepper 3	18/6	B DIR	B PUL	B EN	GND	AL	*spare*
Limits	22/6	12V	X	Y	Z	A	B
Probes	22/6	5V	3D	TS	TP	Mist	*spare*

`B` rows are the provisioned 5th-axis / rotary lines. `3D` = 3D probe, `TS` = toolsetter, `TP` = touch plate. The 12 V fan is fed from the panel, not the probes socket; each motor's 48 V and the common ground are distributed separately.



Harness collapse: many device wires → shared GND, wire-OR'd alarm and a 48 V block → five 1×6 sockets into the controller.



Back-panel layout: the five 1×6 sockets plus the 48 V block, common-ground bus and the USB/Ethernet/C20 pass-throughs.

Overhead umbilical & dust collection

The spindle's services arrive from **above** rather than through a drag chain, on a **ceiling-mounted spring retraction reel** that takes up slack as the gantry moves.

- **Umbilical:** the two 8 mm coolant tubes and a 1/4″ compressed-air tube run from the top of the Z gantry into a **1″ cable sleeve**.
- **Retraction reel:** the sleeve hangs from a spring-loaded retraction reel on the ceiling above the machine. A **3D-printed cradle** holds the sleeve and clips to the reel hook, so the reel's pull-wire can't wrap and choke the sleeve while still allowing free movement.
- **Routing:** the sleeve terminates at the **back of the CNC**, where the coolant tubes join the pump outlet and the radiator inlet (radiator outlet returns to the reservoir). The air tube comes from the pressure-regulator outlet, whose inlet is fed by the 12 V NC Mist solenoid.
- **Dust collection:** alongside the umbilical, a **4″ duct** ends in a **4″ swivel** feeding **5′ of 4″ flex hose** to a **dust shoe** on the spindle.

Rack-and-pinion drive — setup & maintenance

X (and the Y/A gantry drives) run on rack and pinion. After ~10 hours the X pinion crept **left** under load. A closed-loop motor can't see slip *downstream* of its shaft encoder, so a pinion loosening on the shaft is invisible until a fixed reference (the toolsetter at G59.3) exposes it. With the bores reamed

6→8 mm the set-screw boss is thin and grub-screw grip is the weak point; X slips first because it takes the most direct cutting side-load, but the other reamed pinions carry the same risk.

Pinion-to-shaft — stop the slip

1. Pull the pinion; confirm the grub screw seats on the shaft **flat** (not a round section), and check whether the screw marred the flat or only kissed it.
2. **Degrease** shaft, bore, and screw threads — oil defeats both grip and threadlocker.
3. Use the **longest grub screw** the thinned boss will take, on the flat; add a second screw if there's room.
4. **Threadlocker:** blue (242/243) is normally enough; with the reduced thread engagement here, **red (271)** is justified as insurance against back-out — just note removal then needs heat (~250 °C, easy on a bare pinion).
5. Torque to screw size (M3 ~1.0–1.4, M4 ~2.5–3, M5 ~5–6 N·m); snug, don't strip the hex. Let red cure ~24 h before heavy cuts.

Pinion-to-rack mesh & backlash

1. Loosen the motor mount so the pinion can move toward / away from the rack.
2. Shim **0.05–0.10 mm** (feeler gauge, or ~0.1 mm folded paper) between a pinion tooth and the rack, push the pinion in against the shim, tighten the mount, remove the shim.
3. Check the mesh over the **full travel** — it should stay lightly meshed everywhere. Tight at one end + slack at the other means the rack isn't parallel to the axis; realign the rack first.
4. Confirm full tooth-width engagement (pinion not riding a tooth edge).

Verify

1. By hand: smooth full travel, minimal lash; rock the gantry and feel for play at the pinion.
2. Dial indicator: commanded vs actual on small moves; reverse direction to read backlash (a few thou).
3. Re-run a heavy back-and-forth / cut loop with paint-pen **witness marks** (motor shaft ↔ pinion, and a pinion tooth ↔ a rack tooth) — no separation = fixed.
4. Confirm the `AL → ST0` alarm actually halts grblHAL (force a stall) so a real fault can't pass silently.

If it still creeps

A single grub screw — especially in a reamed, thin-wall boss — is marginal for rack-and-pinion loads. Durable fixes, in order: **proper 8 mm-bore pinions** (full set-screw boss) → **clamp / split-hub pinion** (grips the whole shaft, no set screw) → **keyed shaft**. Given the reamed pinions, the **clamp-hub style is the recommended upgrade** — it ends set-screw slip for good.

Sourcing a replacement pinion

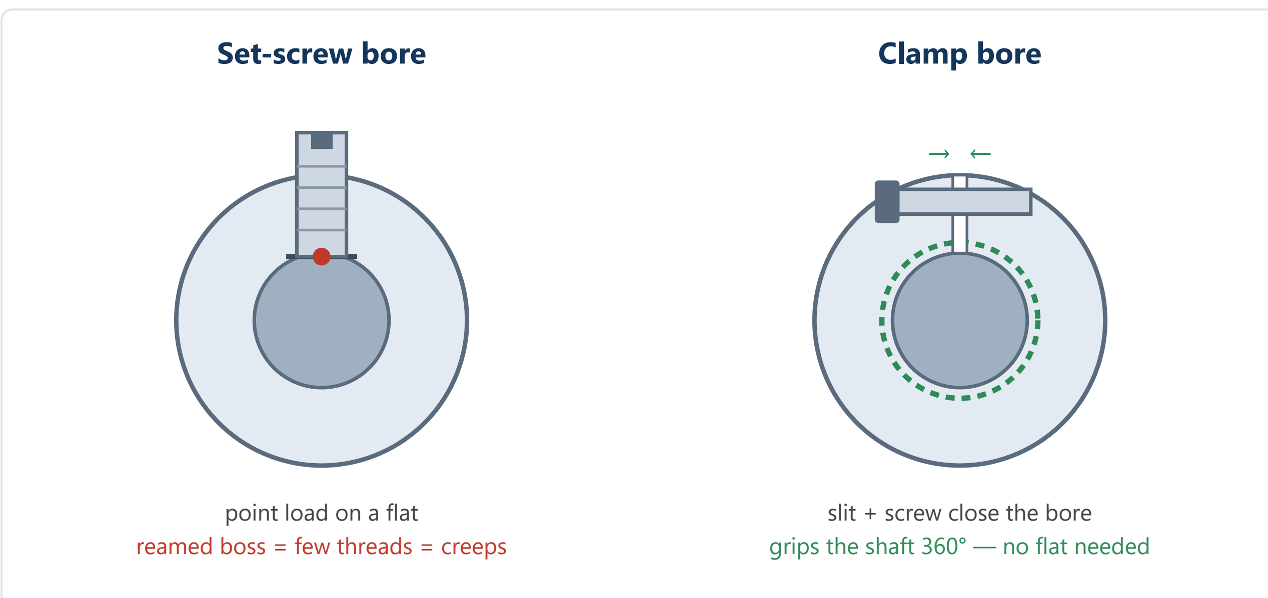
Match the rack on three things, plus the bore:

- **Module** (tooth size) — `module = tooth pitch ÷ π` (pitch 3.14 mm = Mod 1, 4.71 mm = Mod 1.5, 6.28 mm = Mod 2). Must match the rack exactly.
- **Pressure angle** — almost always **20°**.
- **Tooth count** — match the original to keep steps/mm, or pick a new count and recalibrate `$100` (X steps/mm) in grblHAL.
- **Bore** — 8 mm, ideally a **clamp or keyed** bore, not a bare set-screw boss.

Sources that let you specify *module + 20° + 8 mm clamp/keyed bore*: **KHK / QTC** (khkgears.us, qtcgears.com — best for CNC rack & pinion), **SDP/SI** (sdp-si.com), **Misumi** (configurable bore & keyway), and **McMaster-Carr** (quick metric-module steel gears). Budget CNC rack-and-pinion sets are on Amazon / AliExpress (Mod 1/1.5, 20T, often 8 mm bore) with variable quality.

Clamp hub vs set-screw hub

A **clamp (split) hub** grips like a clamp-style shaft collar: a slit is sawn through the bore and a screw bridges it; tightening the screw squeezes the slit closed so the **whole bore clamps the shaft 360°** — far more holding torque, no marring, and no dependence on a thin set-screw boss (exactly the reamed-pinion weakness).



Two routes to a clamp mount: a pinion with an **integral clamp bore** (specify it when ordering from KHK / Misumi / SDP), or a **keyless locking bushing** (Trantorque GT, Fenner B-LOC, …) dropped inside a larger-bore pinion — the bushing expands to grip both the shaft and the bore, letting you use almost any pinion of the right module on the 8 mm shaft. The latter is handy here, since matching bore *and* gearing was the sticking point.

Bill of materials

Item	Qty	Unit price	Extended	Source
Machine & frame				
Mega V XL (Kickstarter edition) — base machine	0	$0.00	$0.00	
Mark Smith riser kit (+3″ Z clearance)	0	$0.00	$0.00	
Mark Smith Z gantry assembly	0	$0.00	$0.00	
Controller & I/O				
Teensy 4.1 SOM (grblHAL controller)	0	$0.00	$0.00	
Teensy 4.x breakout board (Ethernet option)	0	$0.00	$0.00	
USB-A pass-through, panel mount	0	$0.00	$0.00	
Ethernet pass-through, panel mount	0	$0.00	$0.00	
E-stop button (breaks 220 V)	0	$0.00	$0.00	
Power				
Meanwell 48 V power supply, 800 W	0	$0.00	$0.00	
C20 panel inlet — mains 220 V	0	$0.00	$0.00	
C20 panel inlet — VFD 220 V	0	$0.00	$0.00	
Power relay accessory (110 V pump relay)	0	$0.00	$0.00	Amazon B07Z6R1KP3
Spindle & cooling				
2.2 kW 80 mm water-cooled spindle + 2.2 kW VFD kit (ER20, 24k RPM)	0	$0.00	$0.00	Amazon B073TRP92V
Braking resistor, 150 W	0	$0.00	$0.00	
Submersible pump, 110 V	0	$0.00	$0.00	
Coolant reservoir, 1 gal	0	$0.00	$0.00	
Radiator with integrated 12 V fan(s)	0	$0.00	$0.00	
8 mm clear tubing (25′)	0	$0.00	$0.00	
Coolant mix — antifreeze + distilled water + bleach	0	$0.00	$0.00	

Item	Qty	Unit price	Extended	Source
Motion				
Stepper, 1.2 N·m (Z)	0	$0.00	$0.00	
Stepper, 2.0 N·m (X)	0	$0.00	$0.00	
Stepper, 3.2 N·m (Y / A)	0	$0.00	$0.00	
Drag chain, 2″ (Y-axis upsize)	0	$0.00	$0.00	
Sensing & probing				
Proximity sensors (limits / homing)	0	$0.00	$0.00	
3D mechanical probe	0	$0.00	$0.00	
Toolsetter probe	0	$0.00	$0.00	
Air assist				
1/4″ black tubing (10′)	0	$0.00	$0.00	
Needle valve	0	$0.00	$0.00	
Copper tubing (12″)	0	$0.00	$0.00	
Swivel nozzle, 1/8″	0	$0.00	$0.00	
Solenoid valve, 12 V NC	0	$0.00	$0.00	
Pressure regulator (30 PSI)	0	$0.00	$0.00	
3D-printed nozzle bracket	0	$0.00	$0.00	
Cabling & connectors				
Shielded cable, 18/6 (100′)	0	$0.00	$0.00	
Shielded cable, 22/6 (50′)	0	$0.00	$0.00	
Shielded cable, 22/4 (10′)	0	$0.00	$0.00	
Shielded cable, 22/3 (25′)	0	$0.00	$0.00	
12/3 pigtail (extension cord) — mains in to E-stop (joins the breaker-panel circuit)	0	$0.00	$0.00	
12/3 pigtail (extension cord) — E-stop out to VFD	0	$0.00	$0.00	
Mini-Fit Jr 1×6 connectors (5 mated pairs)	0	$0.00	$0.00	

Item	Qty	Unit price	Extended	Source
Mini-Fit Jr crimp terminals (~60)	0	$0.00	$0.00	
48 V distribution terminal block	0	$0.00	$0.00	
Total			**$0.00**	

Not yet itemized (described in the build above but missing from this BOM):

- **Enclosure** — 2020 extrusion, 1/16″ aluminum sheet (5 sides), 1/8″ plexiglass top.
- **Buck converters** — 12 V 3 A and 5 V 3 A.
- **Ferrules** for the terminal-block connections.
- **Touch plate** (probe input in parallel with the 3D probe).
- **Dust collection** — 4″ duct, 4″ swivel, 5′ of 4″ flex hose, dust shoe.
- **Overhead service rig** — spring retraction reel, 1″ cable sleeve, 3D-printed sleeve cradle.